# BASE CHANGE COMPATIBILITY OF THE KERNEL CORRESPONDENCE ALONG AFFINE, FLAT, AND OPEN MORPHISMS

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*Editorial note.* All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file functors.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

*Editorial note.* Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 0 hits for functors.tex (fu-occupied.json, 2026-10-14); consistent with the duty manager's six-check dispatch.

*Source reference.* Original lemma 0FZD: sources/stacks-chow/tags.txt, source lines 19948–19948. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

*Source reference.* Original reference 02KG: sources/stacks-chow/tags.txt, source lines 3371–3371. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

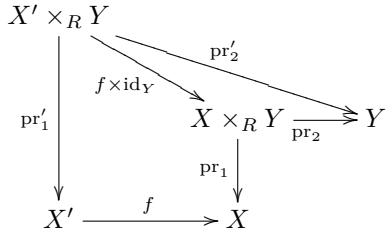
*Source reference.* Original remark 0FZE: sources/stacks-chow/tags.txt, source lines 19949–19949. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

*Source reference.* Original reference 02KH: sources/stacks-chow/tags.txt, source lines 3372–3372. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

**Lemma 0.1.** *In Lemma 0FZD let $F$ correspond to $\mathcal{K}$ in $QCoh(\mathcal{O}_{X\times_R Y})$. We have*

1. *If $f : X' \to X$ is an affine morphism, then $F \circ f_*$ corresponds to $(f \times id_Y)^*\mathcal{K}$.*
2. *If $g : Y' \to Y$ is a flat morphism, then $g^* \circ F$ corresponds to $(id_X \times g)^*\mathcal{K}$.*
3. *If $j : V \to Y$ is an open immersion, then $j^* \circ F$ corresponds to $\mathcal{K}|_{X\times_R V}$.*

*Proof.* Proof of (1). Consider the commutative diagram



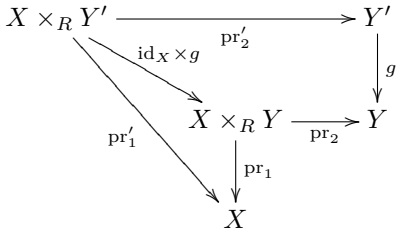

*Date*: stacks-project:functors.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Let $\mathcal{F}'$ be a quasi-coherent module on $X'$. We have

$$\begin{aligned}
\mathrm{pr}_{2,*}(\mathrm{pr}_1^* f_* \mathcal{F}' \otimes_{\mathcal{O}_{X \times_R Y}} \mathcal{K}) &= \mathrm{pr}_{2,*}((f \times \mathrm{id}_Y)_* (\mathrm{pr}_1')^* \mathcal{F}' \otimes_{\mathcal{O}_{X \times_R Y}} \mathcal{K}) \\
&= \mathrm{pr}_{2,*}(f \times \mathrm{id}_Y)_* \left( (\mathrm{pr}_1')^* \mathcal{F}' \otimes_{\mathcal{O}_{X' \times_R Y}} (f \times \mathrm{id}_Y)^* \mathcal{K}) \right) \\
&= \mathrm{pr}'_{2,*}((\mathrm{pr}_1')^* \mathcal{F}' \otimes_{\mathcal{O}_{X' \times_R Y}} (f \times \mathrm{id}_Y)^* \mathcal{K})
\end{aligned}$$

Here the first equality is affine base change for the left hand square in the diagram, see Cohomology of Schemes, Lemma 02KG. The second equality hold by Remark 0FZE. The third equality is functoriality of pushforwards for modules. This proves (1).

Proof of (2). Consider the commutative diagram




We have

$$\begin{aligned}
g^* \mathrm{pr}_{2,*}(\mathrm{pr}_1^* \mathcal{F} \otimes_{\mathcal{O}_{X \times_R Y}} \mathcal{K}) &= \mathrm{pr}'_{2,*}((\mathrm{id}_X \times g)^* (\mathrm{pr}_1^* \mathcal{F} \otimes_{\mathcal{O}_{X \times_R Y}} \mathcal{K})) \\
&= \mathrm{pr}'_{2,*}((\mathrm{pr}_1')^* \mathcal{F} \otimes_{\mathcal{O}_{X \times_R Y'}} (\mathrm{id}_X \times g)^* \mathcal{K})
\end{aligned}$$

The first equality by flat base change for the square in the diagram, see Cohomology of Schemes, Lemma 02KH. The second equality by functoriality of pullback and the fact that a pullback of tensor products it the tensor product of the pullbacks.

Part (3) is a special case of (2). $\square$